PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The impact of virtual care on health-related quality of life in children with diabetes mellitus: A systematic review protocol
AUTHORS	Rajan, Raeesha; Kshatriya, Maya; Banfield, Laura; Athale, Uma; Thabane, Lehana; Samaan, M. Constantine

VERSION 1 – REVIEW

REVIEWER	Brenda Bongaerts Deutsches Diabetes-Zentrum Leibniz-Zentrum fur Diabetes-Forschung, Institute for Biometry and Epidemiology
REVIEW RETURNED	14-Jul-2021

GENERAL COMMENTS	The current manuscript reports the protocol of a planned systematic review into the impact of of virtual care on health-related quality of life in children with diabetes mellitus. It makes perfect sense to me that the authors have published their protocol in Prospero. Hence, it is not clear to me why it is pursued to also publish this protocol in a peer-reviewed journal. This is not a common place to search for systematic reviews being planned/in development. The protocol reads well, the topic is very relevant and up-to-date. The protocol is fairly complete but there are some points that require additional attention. Introduction: • The introduction reads well. Yet I miss somewhat more detailed information on eHealth/mHealth therapy approaches. What is it, how might they work and what experience has been gained thus far with these approached in treatment of diabetes (be it adults or children). To compensate for the extra text, the first two paragraphs on the burden of diabetes in children could be shortened. • The research question does not include all PICO criteria. I´m missing the comparator intervention. Perhaps the type of studies could also be mentioned here, as this was mentioned fairly late in the manuscript. Methods: • It is a very sensible decision to seek help with an experienced librarian for composing a literature search algorithm. However, I do have some comments regarding this algorithm 1. A search algorithm for studies performed in children has been developed by one of the Cochrane review groups. I I would advise to use it. It can be accessed, for example, though a systematic review on children with T1D or T2D. 2. Cochrane also has validated search algorithms for search studies on T2D and T1D. I would also recommend to use these. They can be copied from any recent Cochrane review on T2D and/or T1D. 3. It is not advisable to include search terms for type of studies, in this case randomized studies and observational studies. Study type

	is usually not very well indexed in electronic bibliographies and will therefore restrict the search too much and lead to too few potentially eligible studies. I would advise to delete this part of the search and just include all study types. In the process of study selection, unsuitably designed studies can be easily filtered out. 4. If RCTs are being included, the Cochrane Controlled Register of Trials (CENTRAL) should definitely be searched. 5. Given the topic of the review I would also like to advise to search CINHAL, an electronic bibliography with a nursing and allied health focus. The inclusion criteria: 1. Please define how T1D and T2D should be defined in the studies. 2. Regarding the intervention, please mention whether or not there is a role for parental involvement in the interventions and what the minimum duration of the intervention should be. Are there any exclusion criteria for the intervention to be reported? 3. Regarding the HRQoL measurements, I assume that studies that failed to report one or more outcomes are not excluded for that reason? Outcome measures: 1. A typo appeared in the paragraph on the critical (primary?) outcome: only diabetes-related HRQoL questionnaires are mentioned, while the inclusion criteria also states that general QoL questionnaires are eligible. 2. The other outcomes are clearly described and defined. Outcome 4 on patient and family satisfaction could use some more details/examples. Data Abstraction: 1. Please report in more detail which data will be extracted. What data will be extracted regarding the study populations, regarding the intervention and comparator intervention? What QoL measures will be relevant for extraction, which measures of treatment effect will be used/are relevant? 2. How will be dealt with duplicate and companion publications? Data Analysis: 1. Please state that RCTs will not be pooled with observational data. 2. The fact that a high degree of heterogeneity is to be expected is not a valid reason to resort to a random-effects model. In case of a high levels of heterogeneity, a meta-analysis might not be a good idea. Still, random-effects meta-analysis is a good idea as they assume that underlying study effects follow a normal distribution. 3. Another sentence that is not 100% correct and should be revised (which I, as a methodologist immediately stumbled across) is the one on the I^2. The I^2does not assess heterogeneity. It is merely a measure for the degree of heterogeneity: the percentage of variability in the studies´ effect estimates that are due to heterogeneity, rather than chance. In that respect, are other subgroup analyses being planned to investigate potential sources of heterogeneity, besides stratifying on type of diabetes and age and sex (which are very sensible)? Next to the planned sensitivity analyses, please also consider the exclusion of unpublished studies, or that of long and very large studies (just a suggestion). I wish the authors good luck with their systematic review.

REVIEWER	Helle Haslund-Thomsen Aalborg University Hospital

REVIEW RETURNED	13-Sep-2021

GENERAL COMMENTS	Interesting and relevant protocol. I do have comments on: ….review aims to determine the effectiveness of existing eHealth and mHealth 127 interventions in maintaining or improving HRQOL, glycemic control, overweight and obesity, 128 and adverse events in children living with diabetes. Also, we aim to determine user satisfaction, 129 and their perceptions of the acceptability and feasibility of virtual care modalities. Comment: I think that this is a very broad aim, that could pose problems in the review as there are several outcomes that are in scope in order to determine effectiveness. Could it be more sharp ? Eligible studies which are not published in English will be translated using 156 Google Translate Commment: Google translate is not a reliable tool for translation of whole articles, this opens for misinterpretations and misunderstanding. So either the translation is done by an authorized person or they are excluded.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:

The current manuscript reports the protocol of a planned systematic review into the impact of virtual care on health-related quality of life in children with diabetes mellitus.
It makes perfect sense to me that the authors have published their protocol in Prospero. Hence, it is not clear to me why it is pursued to also publish this protocol in a peer-reviewed journal. This is not a common place to search for systematic reviews being planned/in development.
The protocol reads well, the topic is very relevant and up-to-date. The protocol is fairly complete but there are some points that require additional attention.

- **Response:** We thank the reviewer for their positive comments.

  Protocol papers are a common tool to report the methods used to conduct a systematic review. In fact, when we searched PubMed while we were preparing our response, we identified 1,827 protocols published in the BMJ journals alone.
  We believe that published protocols improve research transparency, mitigate publication bias and selective reporting of results, and help our colleagues and the public know what studies are planned to inform the field and avoid duplication of efforts. While we understand that PROSPERO and other registration tools are one modality to register trials and systematic reviews, it is not mandatory to register studies and many stakeholders, including the public and perhaps journalists, will not routinely search PROSPERO for research activity in each field. Publication of protocols enhances the awareness of the field and the public about research. These are the reasons we are electing to publish our protocol in an open-access journal, in addition to the PROSPERO registration.

Introduction:

1) *The introduction reads well. Yet I miss somewhat more detailed information on eHealth/mHealth therapy approaches. What is it, how might they work and what experience has been gained thus far with these approached in treatment of diabetes (be it adults or children). To compensate for the extra text, the first two paragraphs on the burden of diabetes in children could be shortened.*
   - **Response:** We are grateful to the reviewer for their positive comments. We did add a section in the introduction defining eHealth and mHealth and modified the introduction accordingly (Page 5, Paragraph 3, Line 115-128; Page 6, Paragraph 1, Line 129-134).

     "Virtual care for diabetes during the COVID-19 pandemic is building on existing trends of increased access to remote care using web-based (eHealth) and mobile-based (mHealth) platforms. These platforms included telephone, Skype for Business, and Zoom that allow audio and video communications between patients, families, and healthcare providers (17, 19). The online platforms of continuous glucose monitoring devices, such as Dexcom CLARITY® and FreeStyle® Libre were leveraged for sharing accurate glycemic control and management data with healthcare providers (17, 18, 20, 21).
     eHealth refers to internet, virtual reality, or digital gaming applications that allow users to monitor, manage, or learn more about their health through video, text, or interactive learning media (22-24). mHealth refers to portable or wireless applications such as text messaging, mobile-compatible applications (apps), wearable devices, or the use of social media (22, 23, 25). The integration of virtual care modalities into healthcare is quite promising due to their relatively low cost, scalability, and their association with successful patient engagement, satisfaction with communication with healthcare teams, and the facilitation of access to specialized care and diabetes monitoring (26, 27). "

2) *The research question does not include all PICO criteria. I´m missing the comparator intervention. Perhaps the type of studies could also be mentioned here, as this was mentioned fairly late in the manuscript.*
   - **Response:** We thank the reviewer for this comment. We have clarified the research questions for this systematic review (Page 7, Paragraph 2, Line 155-160).

     "Research questions
     Primary: In children with diabetes, does the use of eHealth and mHealth interventions, when compared to in-person care, maintain or improve HRQOL?
     Secondary: In children with diabetes, do virtual interventions maintain or improve glycemic control, body mass, and adverse diabetes-related outcomes when compared to in-person care? Is virtual care usable and acceptable for this population?"

Methods:

It is a very sensible decision to seek help with an experienced librarian for composing a literature search algorithm. However, I do have some comments regarding this algorithm

1) *A search algorithm for studies performed in children has been developed by one of the Cochrane review groups. I would advise to use it. It can be accessed, for example, though a systematic review on children with T1D or T2D.*
   - **Response:** We have reviewed the language used for children in multiple Cochrane reviews against our search and specific population. The language includes infants and newborns; both groups fall outside the scope of our study population. In addition, like the Cochrane reviews, we used the controlled vocabulary for children and adolescents. These terms are check-tags within the databases and are automatically applied to article records, even before the regular controlled vocabulary. We have determined that our language is more targeted to our population (2-18 years old).

2) *Cochrane also has validated search algorithms for search studies on T2D and T1D. I would also recommend to use these. They can be copied from any recent Cochrane review on T2D and/or T1D.*
   - **Response:** Thank you for your comments. We reviewed the Diabetes search terminology used in recent Cochrane Reviews. We found the language to be consistent, the exception being some Cochrane reviews included Diabetes terms that are part of our exclusion criteria (e.g., MODY). Therefore, we believe that adhering to our current search strategy targets the specific population in this systematic review.

3) *It is not advisable to include search terms for type of studies, in this case randomized studies and observational studies. Study type is usually not very well indexed in electronic bibliographies and will therefore restrict the search too much and lead to too few potentially eligible studies. I would advise to delete this part of the search and just include all study types. In the process of study selection, unsuitably designed studies can be easily filtered out.*
   - **Response:** As recognized by the reviewer's previous comments, Cochrane is considered to be the authoritative source on Systematic Review methodology. We consulted The Cochrane Handbook Chapter 4, section 4.4.2 (The Cochrane Collaboration, 2021); "a search strategy will typically have three sets of terms: (i) terms to search for the health condition of interest, i.e. the population; (ii) terms to search for the intervention(s) evaluated; and **(iii) terms to search for the types of study design to be included**." Thus we incorporated study design into our search strategy. Further, we ran a search without study design and screened a sample of the records that would not have been picked up by the study design language. We were satisfied that we were not missing studies when we include the study design.

4) *If RCTs are being included, the Cochrane Controlled Register of Trials (CENTRAL) should definitely be searched.*
   - **Response:** Thank you for your comments. We did run a test search in CENTRAL, upon review of the results we did not identify any additional studies.

5) *Given the topic of the review I would also like to advise to search CINHAL, an electronic bibliography with a nursing and allied health focus.*
   - **Response:** Thank you for your comments. We did include a nursing and allied health focused database, OVID EMCare, the scope of which is larger than CINHAL. EMCare in comparison to CINAHL indexes 3,500 peer-reviewed journals as opposed to 1,380 for CINAHL. The use of EMCare should cover all CINAHL content and goes beyond it.

The inclusion criteria:

1) *Please define how T1D and T2D should be defined in the studies.*
   - **Response:** Thank you for this suggestion. We have added details on how T1D and T2D should be diagnosed to the inclusion criteria (Page 10, Paragraph 1, Line 181-184)

     "The diagnosis of diabetes is stablished using standardized criteria, with a random or a 2-hour post oral glucose tolerance test ≥11.1 mmol/L (200 mg/dL), or a fasting glucose level of ≥7.0 mmol/L (126 mg/dL) (31)."

2) *Regarding the intervention, please mention whether or not there is a role for parental involvement in the interventions and what the minimum duration of the intervention should be. Are there any exclusion criteria for the intervention to be reported?*
   - **Response:** Thank you for your suggestion. Interventions that involve parent/caregiver participation will be included. This clarification has been added (Page 11, Paragraph 2, Line 202-203)
   - As we are unsure of what we will find, we can't commit to a duration beyond which we will exclude or include an intervention. We will make comments on this issue in the systematic review paper.

3) *Regarding the HRQoL measurements, I assume that studies that failed to report one or more outcomes are not excluded for that reason?*
   - **Response:** This is correct. We have added a clarifying statement to address this comment (Page 12, Paragraph 1, Line 221-222).

Outcome measures:

1) *A typo appeared in the paragraph on the critical (primary?) outcome: only diabetes-related HRQoL questionnaires are mentioned, while the inclusion criteria also states that general QoL questionnaires are eligible.*
   - **Response:** Thank you for pointing this out. We have made the clarification that HRQoL questionnaires and proxy formats will be accepted to the critical outcomes section (Page 12, Paragraph 3, Line 232-233).

2) *The other outcomes are clearly described and defined. Outcome 4 on patient and family satisfaction could use some more details/examples.*
   - **Response:** We have included additional information and examples to the patient and family satisfaction outcome (Page 13, Paragraph 2, Line 254-257).

     "4) Patient and family satisfaction, and their perceptions of the usability and acceptability of the virtual intervention from relevant questionnaires, which include measures to assess frequency and ease of use and desire for continued application after the study has ended."

Data Abstraction:

1) *Please report in more detail which data will be extracted. What data will be extracted regarding the study populations, regarding the intervention and comparator intervention? What QoL measures will be relevant for extraction, which measures of treatment effect will be used/are relevant?*
   - **Response**: We have added more detail on the data that will be abstracted to the data abstraction section (Page 14, Paragraph 2 Line 272-283).

     "We will abstract including first author's name, country, year, number of participants, age, sex, diabetes type, diabetes duration, comorbidities, insulin type and mode of delivery, and modality of glucose monitoring. Data on hyperglycemia, hypoglycemia, ketosis, and ketoacidosis will also be abstracted. We will abstract relevant data on the interventions including type, duration, details of delivery, HRQOL questionnaires used, HRQOL scores, glycemic control measures, change in glycemic control, change in BMI z-score or BMI percentile, adverse events, data on user satisfaction and the usability and acceptability of the interventions. Based on the data available, we will determine whether studies can be pooled for meta-analyses."

2) *How will be dealt with duplicate and companion publications?*
   - **Response:** Duplicates will be removed in Covidence. This data management question is mentioned in the Data Management section of the manuscript (Page 13, Paragraph 3, Line 259).

Data Analysis:

1) *Please state that RCTs will not be pooled with observational data.*
   - **Response:** We have included this statement (Page 15, Paragraph 3, Line 301-302).

2) *The fact that a high degree of heterogeneity is to be expected is not a valid reason to resort to a random-effects model. In case of a high levels of heterogeneity, a meta-analysis might not be a good idea. Still, random-effects meta-analysis is a good idea as they assume that underlying study effects follow a normal distribution.*
   - **Response:** We thank the reviewer for this comment. We have specified the use of a random-effects model for its advantage in assuming normal distribution of the data (Page 15, Paragraph 4, Line 306-307).

3) *Another sentence that is not 100% correct and should be revised (which I, as a methodologist immediately stumbled across) is the one on the I^2. The I^2does not assess heterogeneity. It is merely a measure for the degree of heterogeneity: the percentage of variability in the studies´ effect estimates that are due to heterogeneity, rather than chance. In that respect, are other subgroup analyses being planned to investigate potential sources of heterogeneity, besides stratifying on type of diabetes and age and sex (which are very sensible)? Next to the planned sensitivity analyses, please also consider the exclusion of unpublished studies, or that of long and very large studies (just a suggestion).*
   - **Response:** We accept the reviewer's concern about the description of the I^2 and have revised the manuscript accordingly. This clarification has been made (Page 15, Paragraph 5, Line 310-311 and Page 16, Paragraph 1, Line 312)

     "Heterogeneity will be assessed using the χ2 test p-value with a cut-off value of 0.1 and the I2 statistic with a cut-off value of >75% defined as considerable heterogeneity, in addition to visual assessment of forest plots (50)."

   - We cannot commit to sensitivity analyses that will exclude the above categories, as there may be data that are important to include in the analysis. However, we take the valid reviewer's comment on this issue, and will address it in the paper.

I wish the authors good luck with their systematic review.

- **Response:** Thank you very much for your kind words of encouragement. We are appreciative of the above suggestions that made this manuscript stronger.

Reviewer 2:

Interesting and relevant protocol. I do have comments on:

"…review aims to determine the effectiveness of existing eHealth and mHealth interventions in

maintaining or improving HRQOL, glycemic control, overweight and obesity, and adverse events in children living with diabetes. Also, we aim to determine user satisfaction, and their perceptions of the acceptability and feasibility of virtual care modalities."

I think that this is a very broad aim, that could pose problems in the review as there are several outcomes that are in scope in order to determine effectiveness.

Could it be more sharp ?

- **Response:** We would like to thank the reviewer for their comment. The aims are reported initially in more broad terms that is then further distilled to reflect specific outcome measures. We have added clarification to the research questions that addresses the reviewer's comment (Page 7, Paragraph 2, Line 155-160).

"Research questions
Primary: In children with diabetes, does the use of eHealth and mHealth interventions, when compared to in-person care, maintain or improve HRQOL?
Secondary: In children with diabetes, do virtual interventions maintain or improve glycemic control, body mass, and adverse diabetes-related outcomes when compared to in-person care? Is virtual care usable and acceptable for this population?"

"Eligible studies which are not published in English will be translated using 156 Google Translate." Google translate is not a reliable tool for translation of whole articles, this opens for misinterpretations and misunderstanding. So either the translation is done by an authorized person or they are excluded.

- **Response:** We thank the reviewer for this comment. As we do not know what languages we will encounter based on eligible papers for inclusion in the review the hiring of authorized interpreters for translation is not feasible in terms of cost as there is no specific funding available for this systematic review to allow this step to occur.
- Importantly, we have experience with using Google translate in previous systematic review work (Bhatt, 2018). There are also studies that have assessed the accuracy of Google Translate as a tool for translating papers for use in systematic reviews:
    - Aiken et al.(2011) examined the accuracy of Google Translate, reporting acceptable scores (>70 out of 100) for Portuguese, Italian, Greek, Dutch, and Slovenian. Lower scores (<50 out of 100) were calculated for papers in Japanese, Latvian, Lithuanian, and Polish, and poor scores (0 out of 100) were calculated for documents in Thai.
    - Balk et al (2012) examined the use of Google Translate for data abstraction for systematic reviews and reported good agreement between reviewers for Portuguese and fair agreement between reviewers for Japanese papers in their native languages.
    - Jackson et al (2019) reported high reviewer agreement (85-97%) using Google Translate for Chinese, French, German, Italian, Japanese, Korean, Romanian, Russian, and Spanish.

  Current evidence, including our prior experience, suggests that Google Translate can be reliable in providing translations for the purposes of data abstraction in systematic reviews. In the event we have a paper written in a rare language that Google Translate cannot accurately translate, we will likely exclude the paper and report this exclusion in the full systematic review.

VERSION 2 – REVIEW

REVIEWER	Helle Haslund-Thomsen Aalborg University Hospital
REVIEW RETURNED	22-Nov-2021

GENERAL COMMENTS	The paper is much improved and I have no further comments.